S01 Small-Array dToF
Obstacle Avoidance Radar

2026-10-15 V2.6

MRDVS®
Helping Robots Understand the World



Note: The product shown is the Standard Version. An Industrial Version is also available.

MRDVS S01 is a compact, small-array dToF radar combining cost effectiveness with reliable adaptability to application environments. It supports presence and position detection, short-range blind-spot obstacle avoidance, and height positioning, providing affordable, convenient and reliable 3D environmental perception for a wide range of intelligent mobile devices.

Current Challenges

- During pallet pickup, fork tips are a visual blind spot on autonomous forklifts. Conventional scanners mainly scan around the vehicle and may miss low obstacles or debris between, ahead of or below the forks.
- Lawn mowers have blind spots to the sides and below. Nearby shrubs and flower-bed edges may go undetected, causing missed mowing or collisions.
- Cleaning robots have blind spots and limited detection of materials such as glass. Ranging may fail, delaying detection of low or transparent obstacles.

Obstacle Avoidance Solution

- S01 cameras can be mounted at the tips of both forklift forks, or on lawn mowers and cleaning robots. Serial, I/O or virtual Ethernet signals connect to the host controller, providing stable and efficient obstacle-avoidance data.

Applications

Between-Fork Obstacle Avoidance

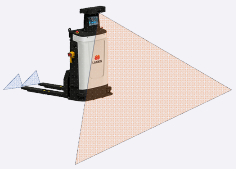
Blind-spot coverage between the forks helps prevent personnel injuries and collision-related stops involving racks, pallets or walls during travel, stacking and turns.

Robotic Lawn Mowers

Side blind-spot obstacle avoidance for safer lawn-mower operation.

Cleaning Robots

Side blind-spot obstacle avoidance for cleaning robots; glass obstacles can be detected in specific scenarios.





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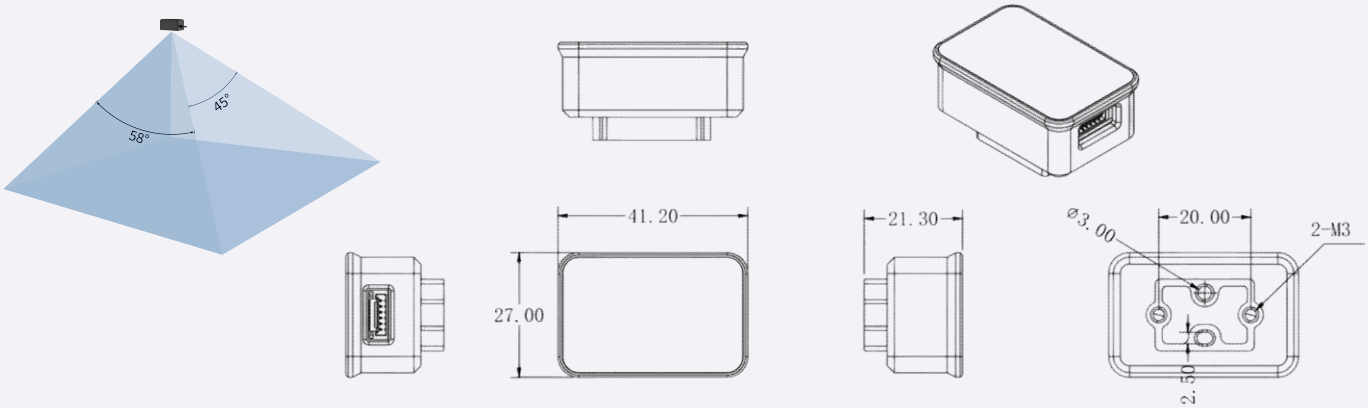
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Product Specifications

Product Model	S01 Small-Array Obstacle Avoidance Radar
Operating Principle	dToF
Output Format	Depth Image, Intensity Image, 3D Point Cloud
ToF Resolution / Frame Rate	40 × 30 @ 10 fps; max. 20 fps
ToF Field of View	58° × 45° (±2°)
Ranging Range	0.15-2 m @ 10%-90% reflectance [1]
Ranging Precision	1σ < 2 cm + 1‰d at 100 klux (80% image area)
Average Operating Power	<1 W
Peak Power	Approx. 4 W
Average Standby Power	<1 W
Laser Wavelength	940 nm
Dimensions	41 × 27 × 21 mm [2]
Weight	Approx. 25 g
Power Supply	Standard Version: 5 V Industrial Version: 12-24 V
Communication Interface	USB [3]
Ingress Protection	IP54
Operating Temperature	-20°C to 50°C
Storage Temperature	-20°C to 60°C
Operating Humidity	0%-95% RH, non-condensing
Ambient Sunlight Immunity	100 klux

[1] Ranging distance is the absolute distance from the optical center.
[2] Dimensions shown are for the Standard Version. The Industrial Version differs slightly; actual dimensions prevail.
[3] CAN and I/O interfaces are supported through expansion. Contact the MRDVS FAE team for details.



Note: Shape and dimensions shown are for the Standard Version; the Industrial Version may differ.